BRIDGES IN MATHEMATICS • GRADE 1 • UNIT 4

Leapfrogs on the Number Line: Skip Counting

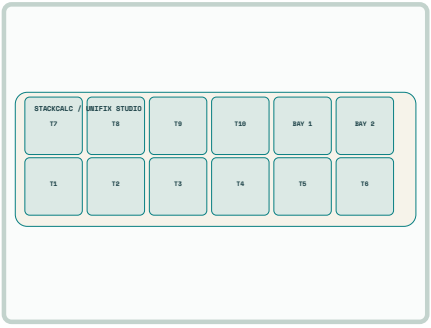
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.1.NBT.A.1

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Unifix Tray

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Arrange Counting & Array tiles into groups of 2, 5, or 10 along the number line.

Sketch Model on Paper:
Draw an open number line with curved hop arcs labeled +2 or +5.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE:	123	Base-10 Arithmetic	
1.	5	X: 5	Hop size 5 into X
2.	ENTER	Y: 5 | X: 5	Stack lift
3.	5	Y: 5 | X: 5	Second hop
4.	+	X: 10	1st sum: 10

EXPECTED LCD X: 20 (4 hops of 5)

1 **Hop 3 times by 2s starting at 0.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Hop 4 times by 5s starting at 0.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Hop 3 times by 10s starting at 0.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **You are at 16. Hop backwards by 2.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Unit 4

Leapfrogs on the Number Line: Skip Counting • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Counting & Array Studio (COUNT-01)

CAD Blueprint: unifix_tray

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/counting_array_studio_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Oceanic Teal Case & 4-Color Array Tiles (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Hop 3 times by 2s starting at 0.** Standard: CCSS.MATH.1.NBT.A.1	6	[2] [ENTER] [2] [+] [2] [+]
2	**Hop 4 times by 5s starting at 0.** Standard: CCSS.MATH.1.NBT.A.1	20	[5] [ENTER] [5] [+] [5] [+] [5] [+]
3	**Hop 3 times by 10s starting at 0.** Standard: CCSS.MATH.1.NBT.A.1	30	[1][0] [ENTER] [1][0] [+] [1][0] [+]
4	**You are at 16. Hop backwards by 2.** Standard: CCSS.MATH.1.NBT.A.1	14	[1][6] [ENTER] [2] [-]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Ensure students say the number sequence aloud in rhythm with stack addition.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.